SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short **Answer Test**

COURSE NAME:

Cloud Computing and COURSE CODE: Big Data Analytics

CSA15

COURSE OUTCOME **COVERED**

CO1: *Apply* cloud computing technologies including *hardware*-software evolution, virtualization, web services, and service models *to* develop cloud-*based* solutions. (BL2)

Dave's Taxonomy: Imitation (Level 1)

Total Marks: 25 Question Count: 5

Code of Conduct

I

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(Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment **Tasks**

Short Answer Test

V. Anish

1.Define cloud computing and explain any four characteristics that make it different from traditional on-

premise computing.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

3.Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

4. Discuss the role of data centers in delivering cloud services..

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short **Answer Test** Rubric

Criteria	**Weightage**	Level **4-** Exemplary	Level 3– Proficient	Level 2- Developing	Level 1- Beginning
Conceptual Accuracy	30%	precise explanation of all	Accurate definitions and	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-	key terms			

points

Covers most expected points

Covers only basic points

Very limited coverage

completely

Uses highly relevant cloud

Use of Examples

15%

examples to strengthen the answer

Uses relevant examples with minor mismatch

Uses vague or partially relevant examples

No useful examples

Comparison/ Differentiation

20%

Clearly distinguishes concepts with strong logic

Clear, organized, and concise answer writing

Clarity of Expression

15%

Reasonable differentiation with minor overlap

Generally clear with few issues

Comparison is weak or incomplete

Understandable but poorly organized

Fails to differentiate concepts

Difficult to follow

Total Marks:

Col-At-1.

Define Clouud computing and Explain any four Characteristic.

doud Computing means using Computing services and software fmoough the internet like storage service intead of maintining them On our own computers.

on-demand self-service usests Can access service whenever they nee them. Example:

Creating a virtual machin in Aws

2. Broad network access: cloud

Services

Can

be accessed

through

tablets using the internet.

22

65

The

mobiles, lap tops

and

Example: Google

Drive. & Resource

Pooling: cloud

among many

users.

Service provides share computing resources

Example: Multiple Companies using

сабе

4 Rapid elasticity: Resource can be

demando

Example: An

festival sales.

cloud

data center

increased or decreased!

to

online shopping website adding servers during

Conclusion cloud

and easy

access

Computing Save Cost and Providers flexible

to Computing

resource.

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2 Explain the evolution of cloud computing

Expl

Cloud Computing developed through different stages large and expensive users shared one power powerful

3

- Hard ware evolution forly computers were large

Mainframe computing &

mony users

Computer.

Personal Computers & Computers became smallers and Pfordable

Client Server Computing client received computers across

from servers

Intanet evolution: The internet

the world.

virtualization's one physical server

virtual machines

connected computers acros

could run multiple

- web software & Applications became accessible though web

- browsers

Modern cloud computing computing resources and software become available through the internet.

Conclusion: Hardware development, the interner ard virtualizing helped in the development of modern cloud Computin g

10/10/2026, 09:47

Explain Server virtualization and differenties it from cloud Computi ng

Server Virtualization is a techinology that divides one physical Sewer into multiple virtual machines: machines: Each virtual machine Can run its own. operation System. each mechine

can

Nun its

630

operating System and
application.

Role of virtualization:

PE

improve

hardware

virtuillization

for physical

servers.

reduce the need

If allows multiple os

to

run on

one servers

It helps allocate

re source easily.

Differences!

Virtualization

It is

a

technology

It create virtual machines

- Example: multiple ums

One server.

cloud computing

it is

a senée model

of provides Computing service

through a network

Example :

Renting

AWS.

a server through

onclusion: Virtualization is one

of the

the important

technologies

Used

to

Provide cloud computing

Computing

service

10/10/2026, 09:47

Differer

clo

u-

the role of data centres in

Explain the role of data

A data center is a faculty containing

and networking equipment

Roles of data centers

used

to

Cloud Computing

servers, storage

device

Store and proces data

1 Data Storage! They

of

information

Store large amount

sev

requests

and

2n

2 Dala processing: They processing

application .

3. Networking: They connect

the internet

Le

security & They

Protect

5 Back up 't They maintain

cloud

server to

users through

data using security system.

Copies of important

6. Availability: Back
up

Back up system help reduce service interruptions

They support addition resource when
demand

7 scalability: They support addition

Increases

Example: Google data centers support Gmail and
Google Drive

Differentiate Taas, paas,

Cloud Computing provides

requirement

Saas and xaas

with example

different service based on user

1- Caas) (Infrastructure as a service)

users

rent computing infrastructure and manage their

Operating Systems

and

application s-

Provides virtual servers, storage and

Example: Amazon EC2

2- paas (platform as a service)

networkin g

- It provides platform to develop and deploy application

- Developers need not manage the underlying infrastructure.

Example: Google app engine

3- Saas (Software as *a* service)

- it provides ready to-use software though the internet.

ready-to-use

users Can access it themselves.

Example: Google Docs

Software without installing and maintaining

a xaas (Everything as service)

. It provides different IT resources

internet.

It includes services such as Storage

Examples Microsoft Azure Services.

10/10/2026, 09:48

and service over the

database and security

10/10/26